

# Compositional monitoring of energy-mix drift on the simplex

*Which carrier did the structural work?*

*Not only which carrier got bigger — which carrier moved the composition.*

*Operationalizing compositional analysis — a runnable standard for researchers and the AI assistants they choose.*

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github.com / PeterHiggins19 / higgins-decomposition

Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

*The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.*

# The size view hides the work

*A carrier can be small in share and large in structural work.*

## World electricity, 25 years — the standard stacked-area view

Coal stays dominant. Gas grows. Nuclear declines.  
Solar appears as a thin yellow sliver after 2010.  
Wind grows steadily but stays visually small until late.

## What the size view misses

### USA Solar, 2012 → 2013

|                        |                      |
|------------------------|----------------------|
| starting share         | 0.107 %              |
| structural Power Share | 81.7 %               |
| Activation Coefficient | $\approx 760 \times$ |

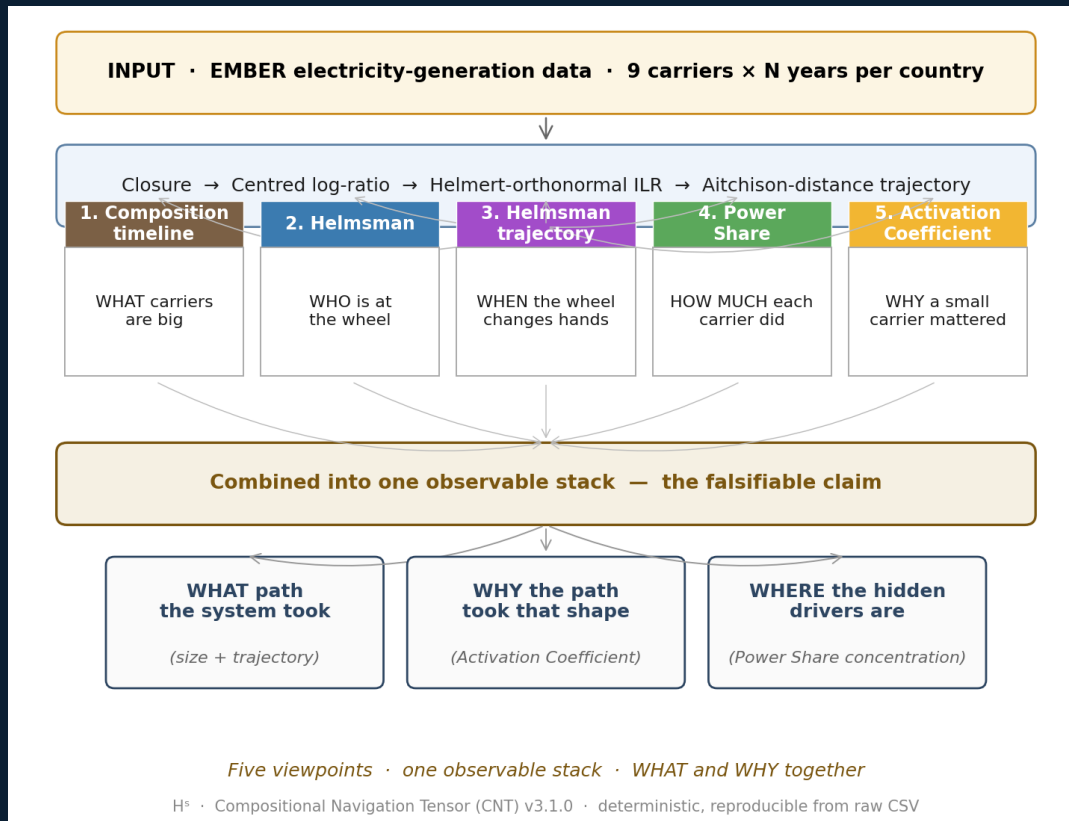
*Solar acted at  $760 \times$  its size.  
No size view shows that.*

*This talk is the reason that number exists.*

*Mathematics: standard compositional data analysis. Application: monitoring frame may be new.*

# Five viewpoints, one observable stack

*Each viewpoint answers one question. Together: an auditable transition event.*



## Composition

what share each carrier has.

## Helmsman

which carrier has the largest CLR move at a step.

## Helmsman trajectory

when the steering carrier changes.

## Power Share

how much squared CLR motion each carrier did.

## Activation Coefficient

Power Share ÷ starting share — the yeast factor.

*All five derive from CLR + ILR-Helmert. Pure CoDa geometry; no new mathematics.*

# The Activation Coefficient — the yeast factor

*How much structural work a carrier does, relative to how much of the mix it is.*

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$  carrier does work proportional to its size — ordinary

$\alpha \gg 1$  carrier acts far above its size — hidden driver

$\alpha < 1$  carrier carries less work than its size suggests — coasting

## Worked example — USA Solar 2012 → 2013

|                |                      |                  |
|----------------|----------------------|------------------|
| starting share | 0.107 %              | small            |
| Power Share    | 81.7 %               | most of the work |
| $\alpha$       | $\approx 760 \times$ | yeast            |

*Yeast is 2% of a loaf by mass and does 100% of the rising. Same shape.*

*Solar 2010–2015 appears repeatedly as small-share / large-structural-work across the corpus.*

# Three archetypes — one instrument, three regimes

*Same protocol applied to three transitions that look fundamentally different.*

## Germany

*deliberate transition*

*continuous arc*

Energiewende  
2000 → 2025  
solar + wind absorb  
structural work  
before size dominates.

## Japan

*external shock*

*loop and reorganise*

Fukushima 2011  
displaces nuclear,  
causes 2011–2013  
multi-year compositional  
reorganisation.

## United Kingdom

*regime change*

*jump and return*

Coal exit 2012–2020  
from > 30 % to < 2 %.  
Wind, solar, others  
absorb displaced  
structural work.

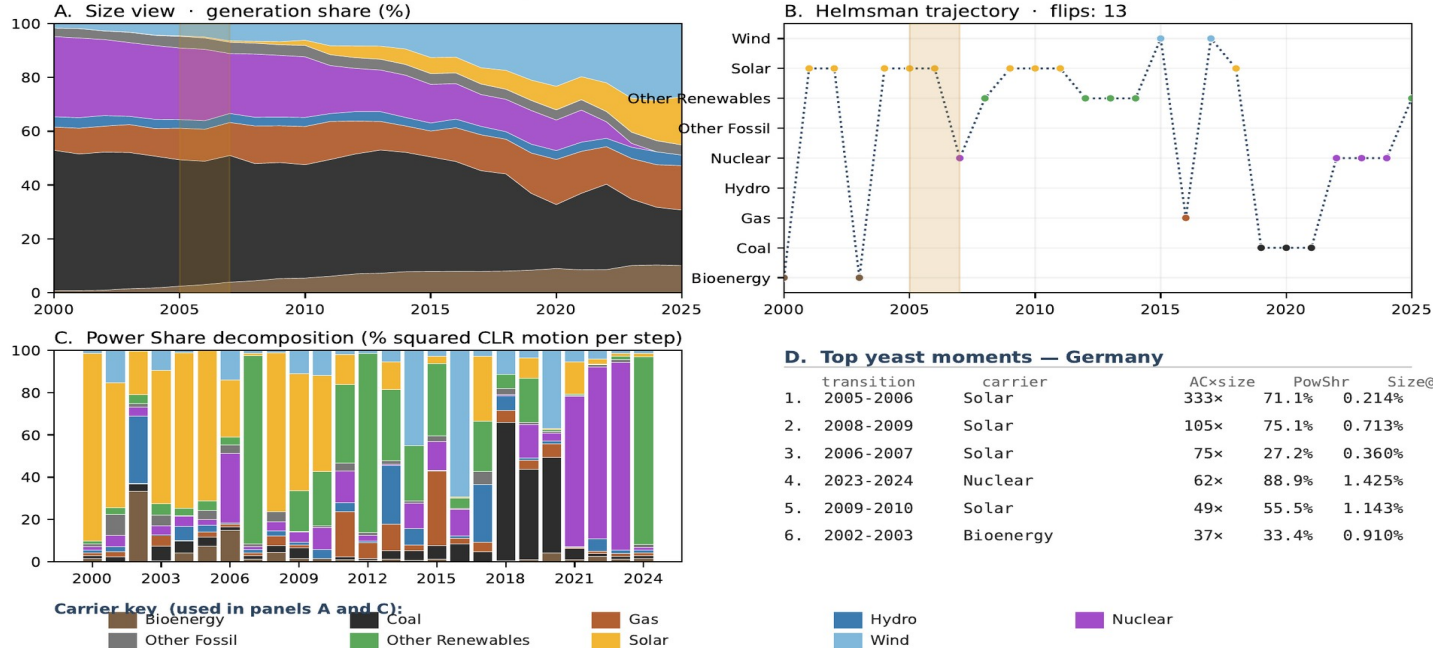
*Three different transition regimes. One operational protocol reads them all.*

# Germany — deliberate transition as continuous course

*The Energiewende read as a single smooth arc on the simplex.*

## Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The size view shift is real — the yeast moments show when it actually accelerated.



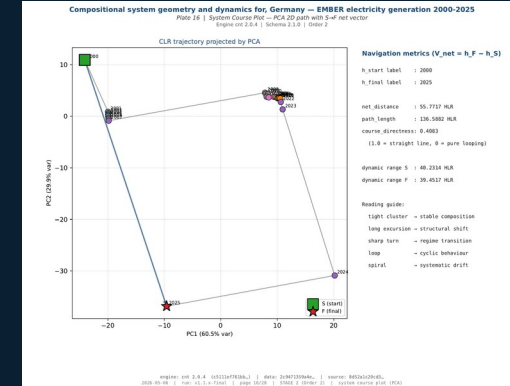
Germany's transition is the textbook case of a continuous arc: coal recedes, gas holds, solar and wind grow steadily.

But the Power Share view shows the acceleration moments — solar in 2005-2006 at 0.21 % composition share doing 71 % of the structural directional work (activation coefficient  $\approx 333\times$ ).

That is the structural beginning of the Energiewende — three years before the size view would call it visible.

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Fig. 2



**Solar 2005-2006: 0.21 % share · 71.1 % structural work ·  $\alpha \approx 333\times$**

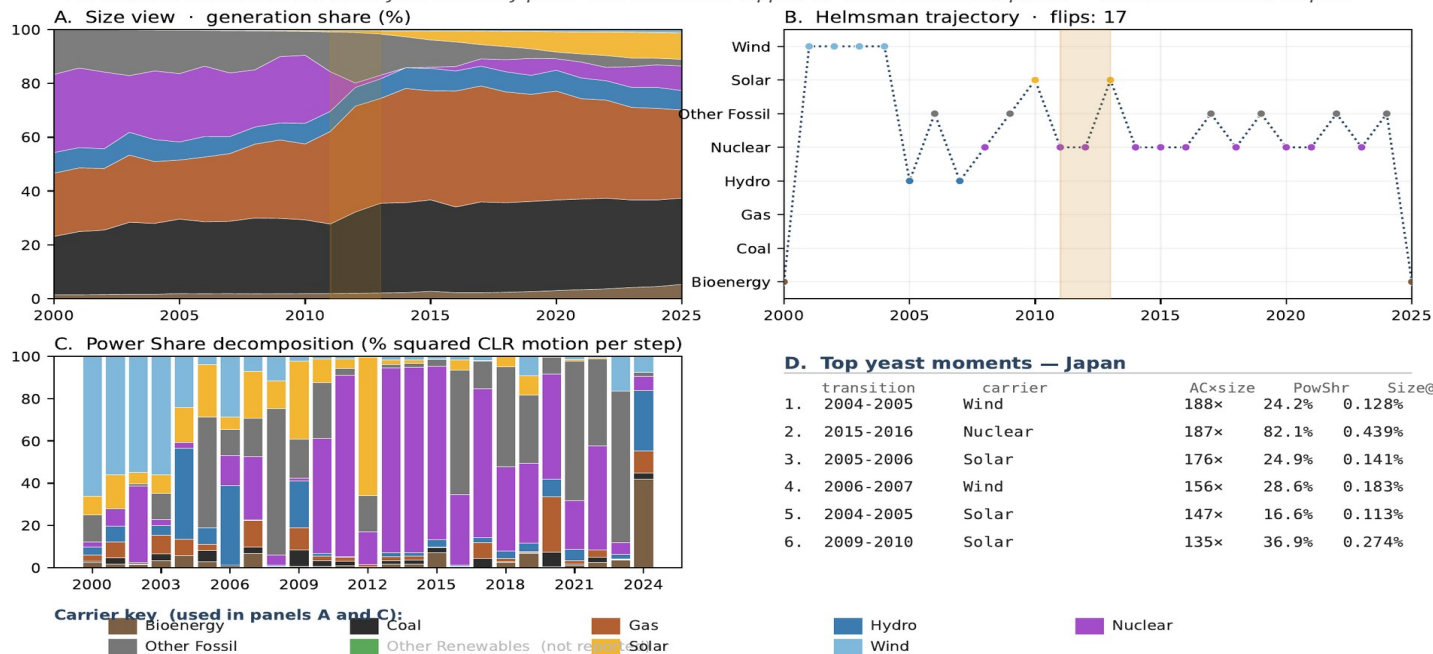
Helmsman trajectory: course directness 0.41 — continuous arc to renewable vertex.

# Japan — Fukushima shock and reorganisation

*The instrument detects both the shock and the multi-year reorganisation.*

## Japan — Fukushima 2011 and the structural aftermath

*An external shock and the multi-year recovery path. The helmsman flipped 17 times across the period — the loudest in the corpus.*

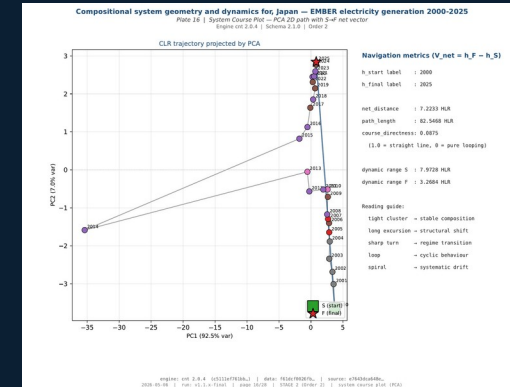


*The 2011-2012 transition shows in every viewpoint at once: composition (nuclear collapses), helmsman (carriers shift abruptly), Power Share (massive concentration of squared CLR motion at one step).*

*The years AFTER tell the deeper story — Japan's post-Fukushima mix kept reorganising for over a decade, and the yeast moments distribute across multiple carriers as solar, gas, and renewables each took turns absorbing the displaced nuclear share.*

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Fig. 3



**Aitchison distance 2011 → 2012 ≈ 3 × neighbouring-year baseline · helmsman flips 17 ×**

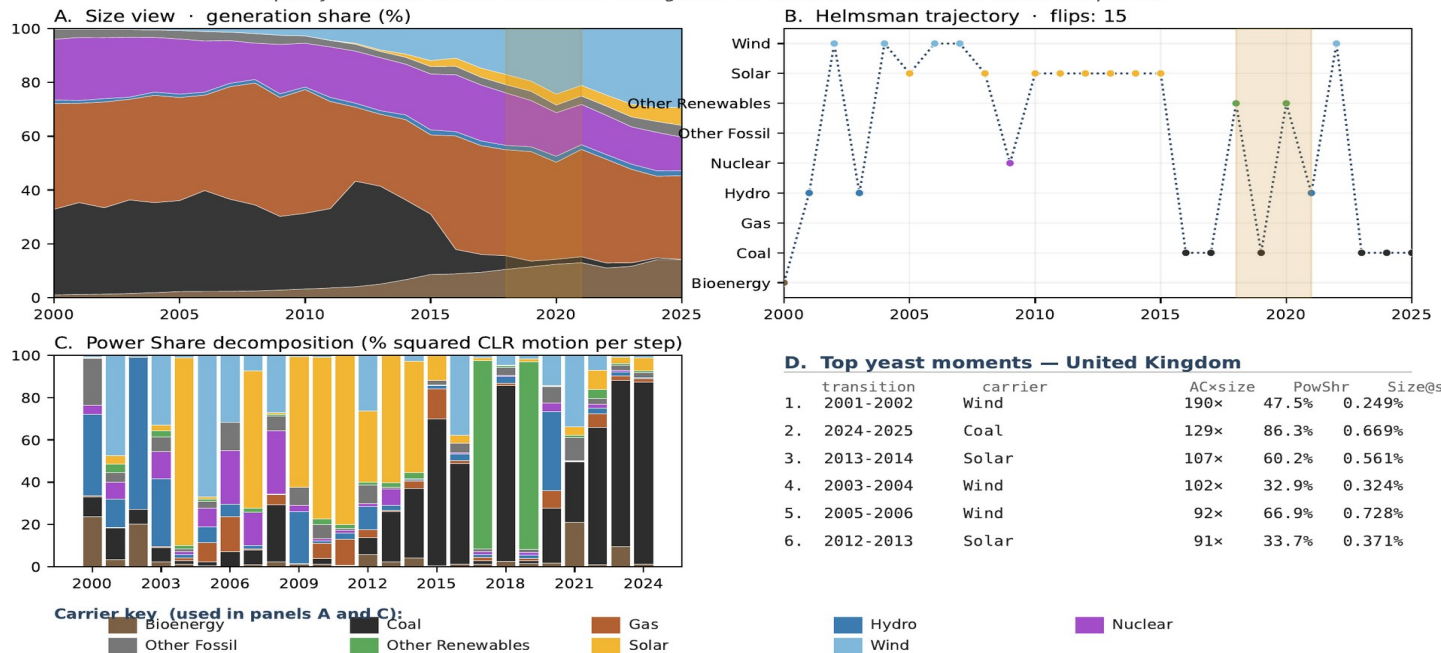
*Course directness 0.09 — looping reorganisation, not a single step.*

# United Kingdom — coal exit as regime change

*Policy-driven displacement, absorbed across multiple renewable carriers.*

## United Kingdom — the coal exit as a regime change

*A policy-driven structural transition. Coal goes from dominant to near-zero across the period.*



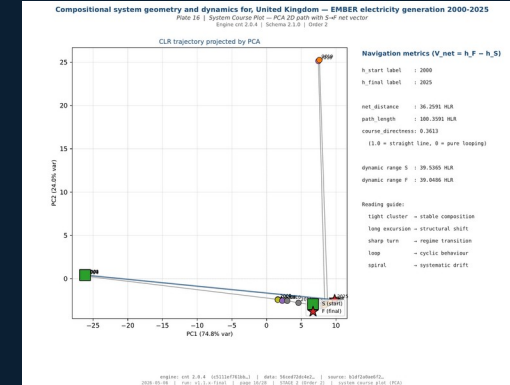
*The UK story is the abrupt-policy archetype. Coal's share drops from over 30 % to under 2 % across the period.*

*The Power Share view names what made it possible: Other Renewables and Wind in the late-2010s took turns doing the structural work of absorbing coal's exit, often at low single-digit composition share (panel D).*

*Without the activation coefficient, those small-share carriers are invisible. With it, the whole transition snaps into focus.*

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Fig. 4



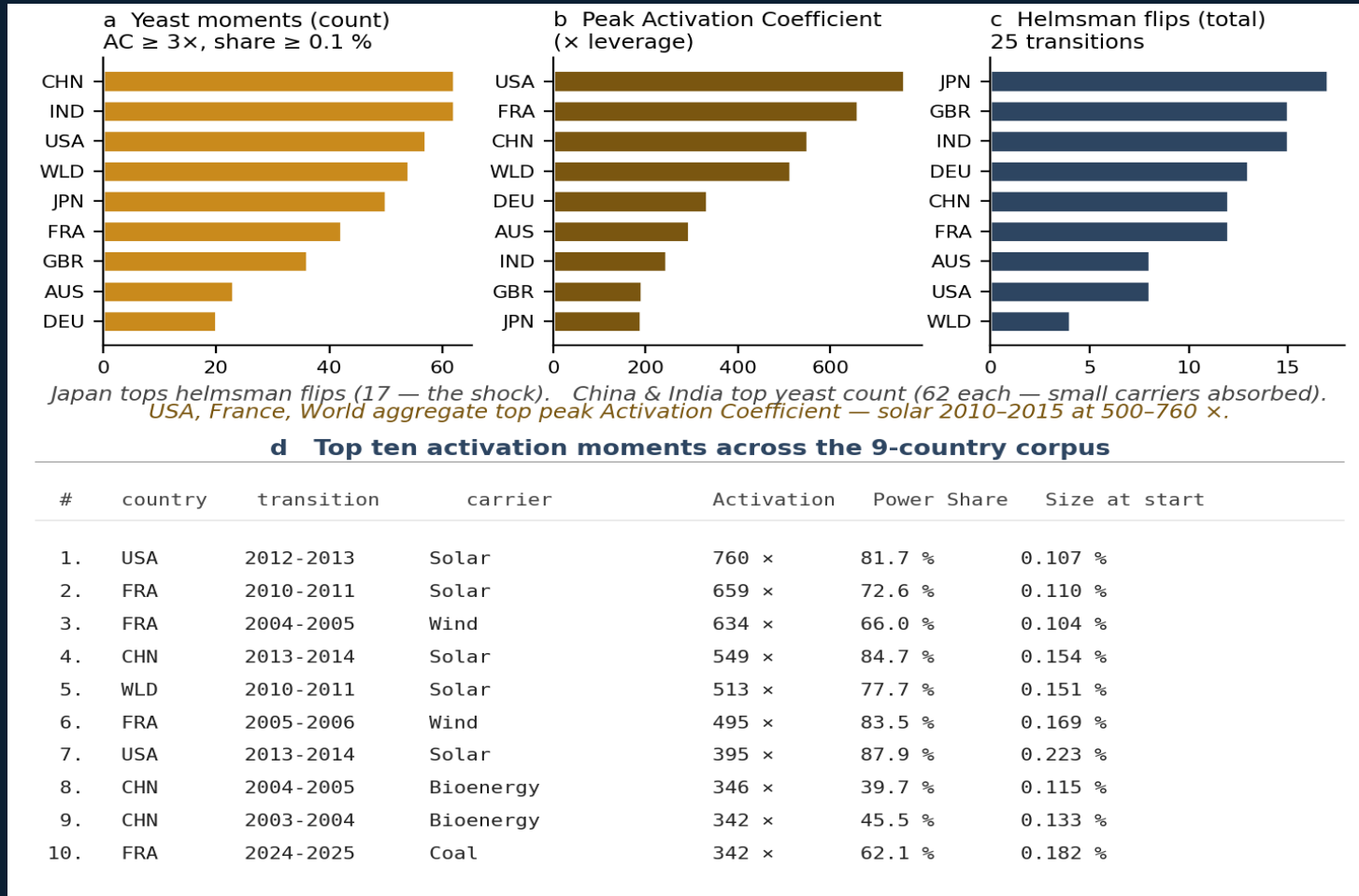
**Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced structural work.**

*Course directness 0.36 — jump-and-return archetype.*



# Cross-country signature — 5 of 9 reproduce the pattern

From three case archetypes to a corpus-level result.



**fires** AUS • CHN • GBR • IND • JPN

**quiet** DEU (annual) • FRA • USA • WLD

A useful detector should not fire everywhere. Discrimination is itself evidence.

# What the stack answers

*One observable, five distinct questions, one reproducible object.*

|                 |                           |                               |
|-----------------|---------------------------|-------------------------------|
| <b>WHAT</b>     | carriers are big.         | <i>size view</i>              |
| <b>WHO</b>      | is at the wheel.          | <i>Helmsman</i>               |
| <b>WHEN</b>     | the steering changes.     | <i>Helmsman trajectory</i>    |
| <b>HOW MUCH</b> | work each carrier did.    | <i>Power Share</i>            |
| <b>WHY</b>      | a small carrier mattered. | <i>Activation Coefficient</i> |

*The stack does not replace interpretation. It gives interpretation a reproducible object.*